

Time & Work

COMPLETE LEARNING GUIDE

What You'll Learn

Time and work problems test the same core insight in progressively disguised forms: work is best measured in RATE (work done per unit time), not in raw days, and rates from different workers combine by simple addition while the DAYS they take do not. The single fastest technique across this entire topic is converting everyone's individual time into a shared 'total work' figure using the LCM of the given times, which turns every worker's contribution into a clean whole-number rate. This sub-module covers the LCM-based rate method, combining multiple workers, workers joining or leaving midway, efficiency ratios, splitting wages by work contribution, the men-days(-hours) formula for comparing different workforce sizes, and alternating-day work schedules — each demonstrated with a complete worked calculation so that recognizing the right setup becomes automatic under time pressure.

1. Work-Rate Basics: The LCM Method

Rather than working with fractions of a day, assign the total job a convenient 'total work' value equal to the LCM of all the individual completion times given in the question — this makes every worker's daily rate a clean whole number, avoiding fraction arithmetic throughout the rest of the problem.

- A can complete a job in 12 days, B in 18 days. Let total work = $\text{LCM}(12,18) = 36$ units. A's rate = $36/12 = 3$ units/day. B's rate = $36/18 = 2$ units/day.

2. Two or More People Working Together

When workers act simultaneously, their rates simply ADD (never their times), and the combined time to finish = total work $\div$ combined rate.

- Using A's rate (3 units/day) and B's rate (2 units/day) from above, working together their combined rate = 5 units/day. Time to finish together = $36/5 = 7.2$ days.

3. Workers Joining or Leaving Midway

When the workforce changes partway through a job, compute how much work was completed before the change, subtract it from the total to find the remaining work, and divide that remaining work by the (new) applicable rate — never apply the original combined time to the whole job.

- A and B (rates 3 and 2 units/day, total work 36 units) work together for 2 days, then B leaves and A finishes alone. Work done in 2 days = $5 \times 2 = 10$ units. Remaining work = $36 - 10 = 26$ units. A alone finishes the remainder in $26/3 \approx 8.67$ days. Total time = $2 + 8.67 \approx 10.67$ days.

4. Efficiency Ratios

Efficiency and time taken are INVERSELY proportional — if worker A is k times as efficient as worker B, A takes only $1/k$ of the time B takes for the same job. This is a common source of error since it's tempting to assume efficiency ratio and time ratio are the same.

- A is twice as efficient as B, and B alone takes 20 days to finish a job. Since efficiency ratio A:B = 2:1, the TIME ratio is the inverse, 1:2 — so A takes $20/2 = 10$ days.

5. Splitting Wages by Work Contribution

When multiple workers complete a job together and are paid a combined amount, the payment must be split in the ratio of their WORK contributed (i.e., their rate ratio), not by the number of days each one worked.

- A can do a job in 10 days, B in 15 days. Working together, they earn ₹5000 total for the job. Rate ratio A:B = $1/10 : 1/15 = 3:2$ (after multiplying both by 30). A's share = $3/5 \times 5000 = ₹3000$. B's share = $2/5 \times 5000 = ₹2000$.

6. The Men-Days(-Hours) Formula

To compare different workforce sizes, durations, or working hours for the SAME total amount of work, use $M \cdot D \cdot (H) = M_2 D_2 (H_2)$, where M is the number of workers, D is the number of days, and H is hours worked per day (included only if given).

- 15 men can build a wall in 20 days. How many men are needed to build it in 12 days? $M \cdot D = M_2 D_2$! $\Rightarrow 15 \times 20 = M_2 \times 12$! $\Rightarrow M_2 = 25$ men.
- 12 men working 8 hours/day finish a job in 10 days. How many days will 8 men working 6 hours/day take for the same job? $M \cdot D \cdot H = M_2 D_2 H_2$! $\Rightarrow 12 \times 10 \times 8 = 8 \times D_2 \times 6$! $\Rightarrow 960 = 48 \times D_2$! $\Rightarrow D_2 = 20$ days

7. Alternating-Day Work Schedules

When workers alternate days rather than working simultaneously, compute the work done per full cycle (e.g., one day each), find how many complete cycles are needed, then handle the final partial day separately by checking whose turn it is.

- A can finish a job in 8 days, B in 12 days. Using total work = $\text{LCM}(8,12) = 24$ units, A's rate = 3/day, B's rate = 2/day. They work on alternate days starting with A. Each 2-day cycle (A then B) completes $3+2 = 5$ units. After 4 full cycles (8 days): 20 units done, 4 units remain. Day 9 is A's turn (odd days are A): A completes 3 units, cumulative 23, 1 unit remains. Day 10 is B's turn: B's rate is 2 units/day, so the last 1 unit takes $1/2$ day. Total time = 9.5 days.

Common Pitfalls

- Adding workers' individual TIMES directly (e.g., averaging 12 and 18 days) instead of adding their RATES (1/time) to find combined performance.
- Forgetting to recompute the remaining work when a worker joins or leaves midway, and instead applying the original combined completion time to the whole job.
- Assuming efficiency ratio and time-taken ratio are the same, instead of recognizing they are inverses of each other.
- Splitting wages by the number of DAYS each worker worked instead of by their actual work-rate (contribution) ratio.
- Forgetting to include the hours-per-day factor in the men-days formula when the question specifies different working hours for each scenario.
- In alternating-day problems, using the wrong worker's rate for the final partial day by not checking whose turn it actually is.
- Using the LCM of the WRONG set of numbers (e.g., only two of three given times) when more than two workers or rates are involved.

Key Takeaways

- Assign total work = LCM of all given individual completion times, so every rate becomes a clean whole number.
- Combined rate = sum of individual rates; time together = total work ÷ combined rate — never add times directly.
- Efficiency and time are inversely proportional: k times as efficient means 1/k the time for the same job.
- Wages are split in the ratio of work contributed (rate ratio), not days worked.
- Men-days(-hours) formula: $M \cdot D \cdot (H \cdot) = M \cdot D \cdot (H \cdot)$ for the same total job.
- For joining/leaving-midway problems, compute work already done first, then divide the REMAINING work by the new applicable rate.
- For alternating-day schedules, compute work per full cycle first, then handle the final partial day by checking whose turn it is.

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GeeksforGeeks — Time and Work Aptitude Questions

<https://www.geeksforgeeks.org/aptitude/time-and-work-questions-and-answers/>

IndiaBix — Time and Work Aptitude Questions

<https://www.indiabix.com/aptitude/time-and-work/>